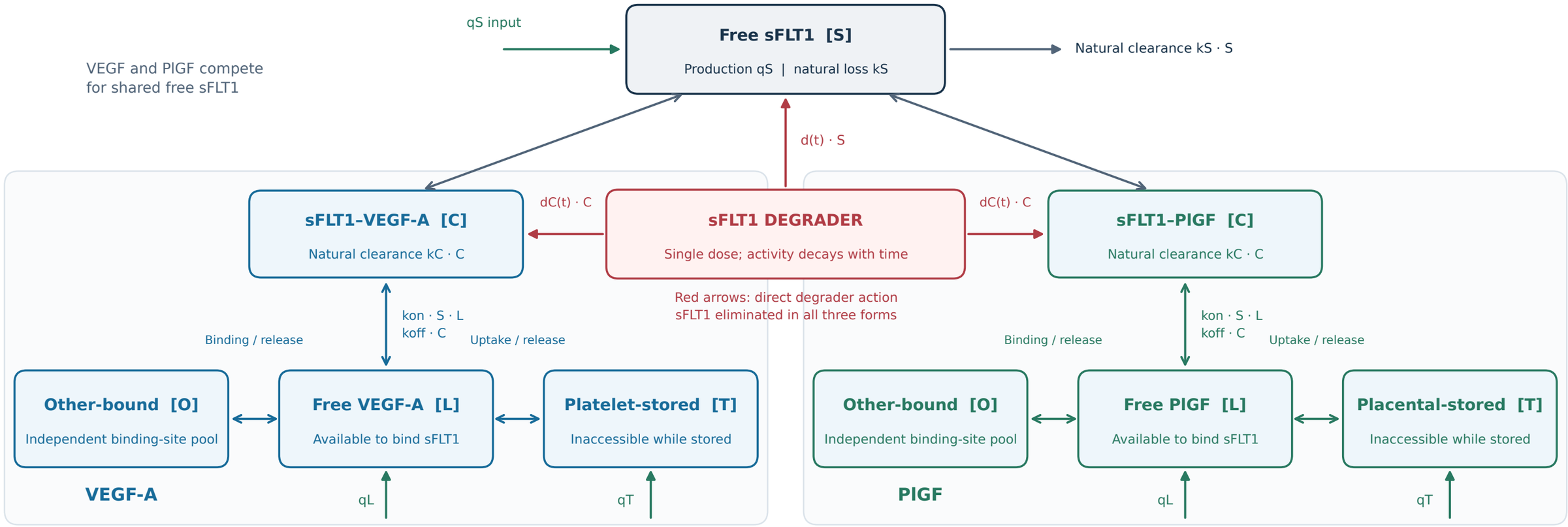
sFLT1 degrader model

VEGF-A and PlGF distribution, reversible storage, and ligand fate during sFLT1 removal



LIGAND FATE DURING COMPLEX DEGRADATION

25% of VEGF and 25% of PlGF are co-degraded; the remaining ligand returns to its free pool. Free sFLT1 removal carries no ligand.

dC(t) = d(t) × relative activity on bound sFLT1 | Other-bound and storage pools have no direct degrader clearance.

Starting concentration (pg/mL)	VEGF-A	PlGF
Free plasma	6.48	33.73
sFLT1-bound / other-bound / stored (each)	31.17	22.09
Plasma total / all-pools total	68.83 / 100.00	77.91 / 100.00
Kd: sFLT1 / other / effective storage (pM)	20 / 20 / 20	150 / 150 / 150

NATURAL LOSS & INPUTS

Every state has first-order natural loss.
Complex loss destroys ligand; other-site capacity is maintained.
qS, qL and qT continue throughout treatment.
Storage also releases at rT (default zero).

Production is calibrated to initial untreated steady state.
Single dose day 0: initial 2/day, half-life 3 days.
Storage and affinity are effective assumptions.

Plasma total = L + C + O; all pools = L + C + O + T. Independent O/T sites for each ligand; only sFLT1 is shared.

Exploratory model, not a validated physiological or assay prediction. Source: model.py defaults; regeneration script: generate_model_schematic.py.